TASK 1 : Develop a C program using fork() to create a **parent and child process** and display:

- PID
- PPID
- Process state

```
krishnasree08@krishnasree08:~$ mkdir week3
mkdir: cannot create directory ‘week3’: File exists
krishnasree08@krishnasree08:~$ cd week3
krishnasree08@krishnasree08:~/week3$ pwd
/home/krishnasree08/week3
krishnasree08@krishnasree08:~/week3$ nano task3.c
krishnasree08@krishnasree08:~/week3$ gcc task3.c -o task3
krishnasree08@krishnasree08:~/week3$ ls
File1.txt  File2.txt  copy.c  copy_program  task1.c  task1_program  task3  task3.c
krishnasree08@krishnasree08:~/week3$ ./task3
Before fork():
PID  = 634
PPID = 422


Parent Process:
PID  = 634
PPID = 422
Child PID = 635
State: Running
Parent waiting for child...
Child Process:
PID  = 635
PPID = 634
State: Running
Child Process: Terminated
Child has terminated.
Parent Process: Running
```

```c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid;

    printf("Before fork():\n");
    printf("PID  = %d\n", getpid());
    printf("PPID = %d\n\n", getppid());

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    else if (pid == 0) {
        // Child process
        printf("Child Process:\n");
        printf("PID  = %d\n", getpid());
        printf("PPID = %d\n", getppid());
        printf("State: Running\n");

        sleep(5);

        printf("Child Process: Terminated\n");
        exit(0);
    }

    else {
        // Parent process
        printf("\nParent Process:\n");
        printf("PID  = %d\n", getpid());
        printf("PPID = %d\n", getppid());
        printf("Child PID = %d\n", pid);
        printf("State: Running\n");

        printf("Parent waiting for child...\n");
        wait(NULL);

        printf("Child has terminated.\n");
        printf("Parent Process: Running\n");
    }

    return 0;
}
```

TASK 2 :

Ready → Running → Waiting → Terminated

using:

- ps
- top
- /proc
-

```
krishnasree08@krishnasree08:~/week3$ nano task2.c
krishnasree08@krishnasree08:~/week3$ gcc task2.c -o task2
krishnasree08@krishnasree08:~/week3$ ./task2 &
[1] 786
krishnasree08@krishnasree08:~/week3$ Process started.
PID: 786
Process is running...
```

```c
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("Process started.\n");
    printf("PID: %d\n", getpid());

    printf("Process is running...\n");

    sleep(20);

    printf("Process completed.\n");

    return 0;
}
```

```
  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    1 root      20   0   21736  13080   9736 S   0.0   0.2   0:01.21 systemd
    2 root      20   0    3180   2200   2072 S   0.0   0.0   0:00.02 init-systemd(Ub
    6 root      20   0    3196   2132   2008 S   0.0   0.0   0:00.01 init
   55 root      19  -1   50444  16200  15104 S   0.0   0.2   0:00.32 systemd-journal
  109 root      20   0   25284   6624   5124 S   0.0   0.1   0:00.56 systemd-udevd
  128 systemd+  20   0   21344  13288  11160 S   0.0   0.2   0:00.21 systemd-resolve
  135 systemd+  20   0   91036   7992   7032 S   0.0   0.1   0:00.16 systemd-timesyn
  197 root      20   0    4244   2684   2432 S   0.0   0.0   0:00.02 cron
  198 message+  20   0    9540   5404   4712 S   0.0   0.1   0:00.10 dbus-daemon
  217 root      20   0   17976   8776   7740 S   0.0   0.1   0:00.16 systemd-logind
  251 syslog    20   0  222516   5916   4536 S   0.0   0.1   0:00.12 rsyslogd
  260 root      20   0    3124   1972   1836 S   0.0   0.0   0:00.01 agetty
  266 root      20   0  107044  23112  13656 S   0.0   0.3   0:00.25 unattended-upgr
  420 root      20   0    3188   1104    980 S   0.0   0.0   0:00.00 SessionLeader
  421 root      20   0    3204   1244   1108 S   0.0   0.0   0:00.07 Relay(422)
  422 krishna+  20   0    6596   5948   3700 S   0.0   0.1   0:00.08 bash
  423 root      20   0    6704   4672   3884 S   0.0   0.1   0:00.02 login
  471 krishna+  20   0   20328  11516   9392 S   0.0   0.2   0:00.22 systemd
  472 krishna+  20   0   21176   3576   1848 S   0.0   0.1   0:00.00 (sd-pam)
  498 krishna+  20   0    6080   5368   3696 S   0.0   0.1   0:00.04 bash
  921 krishna+  20   0    9296   5688   3512 R   0.0   0.1   0:00.00 top
```